

Final Project Report Template

Help Desk Ticket Management System

Section No.	Contents
1	INTRODUCTION
1.1	Project Overview
1.2	Purpose
2	IDEATION PHASE
2.1	Problem Statement
2.2	Empathy Map Canvas
2.3	Brainstorming
3	REQUIREMENT ANALYSIS
3.1	Solution Requirement
3.2	Data Flow Diagram
3.3	Technology Stack
4	PROJECT DESIGN
4.1	Problem Solution Fit
4.2	Proposed Solution
4.3	Solution Architecture
5	PROJECT PLANNING & SCHEDULING
5.1	Project Planning
6	FUNCTIONAL AND PERFORMANCE TESTING
6.1	Performance Testing

7	RESULTS
7.1	Output Screenshots
8	ADVANTAGES & DISADVANTAGES
9	CONCLUSION
10	FUTURE SCOPE
11	APPENDIX
11.1	Source Code
11.2	GitHub Repository Link
11.3	Project Demo Link

1. INTRODUCTION

1.1 Project Overview

The **Help Desk Ticket Management System using ServiceNow** is a cloud-based application developed to simplify and automate the IT support process within an organization. The system enables employees to create support tickets, monitor their status, and receive timely updates. It automatically assigns tickets to support agents using Business Rules and manages workflows through Flow Designer. The application also provides email notifications for critical tickets, along with reports and dashboards for performance monitoring. By centralizing support operations, the system improves efficiency, transparency, and overall service quality.

1.2 Purpose

The purpose of the **Help Desk Ticket Management System** is to provide a centralized platform for managing IT support requests efficiently. It aims to reduce manual effort by automating ticket creation, assignment, and status tracking. The system improves communication between employees and the IT support team through real-time updates and email notifications. It also helps managers monitor support performance using reports and dashboards. Overall, the solution enhances productivity, ensures faster issue resolution, and delivers a better user experience.

2. IDEATION PHASE

2.1 Problem Statement

A problem statement clearly describes the issue faced by the users and explains why a solution is required. It helps identify the root cause of the problem and provides a clear direction for developing an effective solution. A well-defined problem statement ensures that the proposed system addresses the actual needs of the users.

For the **Help Desk Ticket Management System using ServiceNow**, employees experience delays in reporting and resolving IT-related issues due to manual support processes. The absence of a centralized ticket management system leads to inefficient communication, improper ticket assignment, lack of status tracking, and delayed issue resolution. These challenges reduce employee productivity and increase the workload on the IT support team.

The proposed system aims to automate the support ticket lifecycle by enabling employees to create tickets, automatically assigning them to the appropriate support team, providing real-time status tracking, generating notifications for critical tickets, and offering reports and dashboards for better monitoring and decision-making.

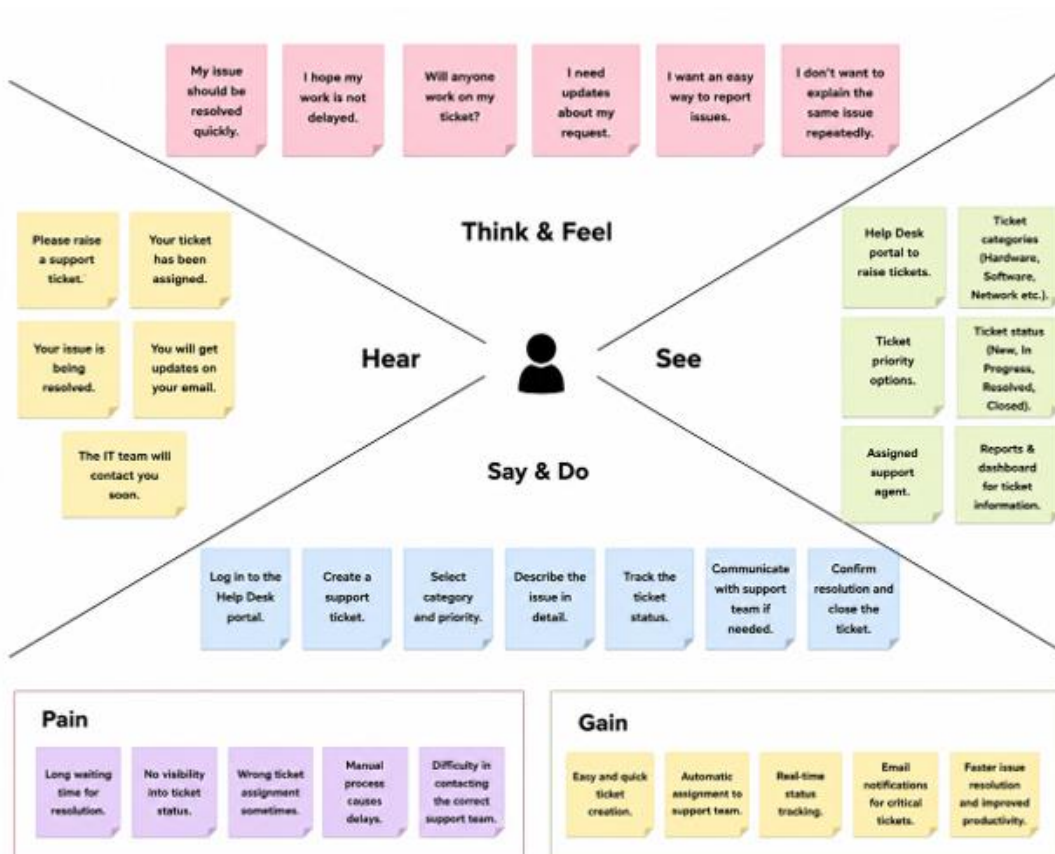
 I AM	Employees of an organization who face IT-related issues and require support from the IT help desk.
 I'M TRYING TO	Raise support requests quickly, get issues resolved, track ticket status, and receive timely updates.
 BUT	Manual ticket handling, no proper tracking, delayed assignment, lack of communication, and long resolution time.
 BECAUSE	Absence of a centralized system, manual processes, lack of automation, and ineffective communication between employees and IT teams.
 WHICH MAKES ME FEEL	Frustrated due to delays, worried about ticket status, uncertain about resolution, and dissatisfied with slow support.

2.2 Empathy Map Canvas

An **Empathy Map Canvas** is a simple and easy-to-understand visual tool that helps capture a user's thoughts, feelings, actions, and experiences. It enables the project team to better understand the needs, expectations, challenges, and behaviors of the end user.

Creating an effective solution requires understanding the real problem from the user's perspective. The empathy mapping process helps identify what users **think, feel, see, hear, say, and do**, along with the **pain points** they experience and the **gains** they expect. This understanding assists in designing a user-centered solution that effectively addresses their requirements.

For the **Help Desk Ticket Management System**, the empathy map focuses on employees who raise support tickets and interact with the IT support team. It highlights their expectations for quick issue resolution, transparent ticket tracking, timely updates, and an efficient support process.



The empathy map helped identify the needs, expectations, challenges, and goals of employees using the Help Desk Ticket Management System. Understanding the user's perspective enabled the development of a solution that provides efficient ticket management, automated workflows, timely notifications, and improved communication between employees and the IT support team.

2.3 Brainstorming

Brainstorming is a creative process used to generate multiple ideas and identify the most suitable solution for a given problem. It encourages the exploration of different approaches before selecting the best one based on feasibility, impact, innovation, and implementation.

For this project, several ideas related to improving organizational support services were considered. Each idea was evaluated based on ease of implementation, scalability, automation capability, and user benefits. After careful analysis, the **Help Desk Ticket Management System using ServiceNow** was selected as the most appropriate solution because it automates the ticket management process, improves communication, and provides efficient tracking and reporting features.

Brainstorming Table

Idea	Advantages	Limitations
Employee Leave Management System	Automates leave requests	Limited to HR operations
Asset Management System	Tracks company assets	Does not address support requests
Visitor Management System	Improves visitor tracking	Limited organizational impact
Help Desk Ticket Management System	Automated ticket assignment, notifications, reporting, dashboards	Requires ServiceNow platform

3. REQUIREMENT ANALYSIS

3.1 Solution Requirement

Functional Requirements

The functional requirements describe the core features and operations of the Help Desk Ticket Management System using ServiceNow.

S.No	Functional Requirement	Description
1	User Authentication	Employees, Support Agents, and Managers can securely log in.
2	Ticket Creation	Employees can create support tickets with category, priority, and description.
3	Ticket Assignment	Tickets are automatically assigned to the appropriate support agent.
4	Ticket Status Tracking	Users can view the current ticket status (New, In Progress, Resolved, Closed).
5	Priority Management	Tickets are categorized as Low, Medium, High, or Critical.
6	Email Notification	Critical tickets trigger automatic email notifications.
7	Ticket Update	Support agents can update ticket status and resolution details.
8	Ticket Closure	Managers can verify and close resolved tickets.
9	Reports	Generate reports based on priority, status, and assigned agents.
10	Dashboard	Display ticket statistics and performance using dashboards.

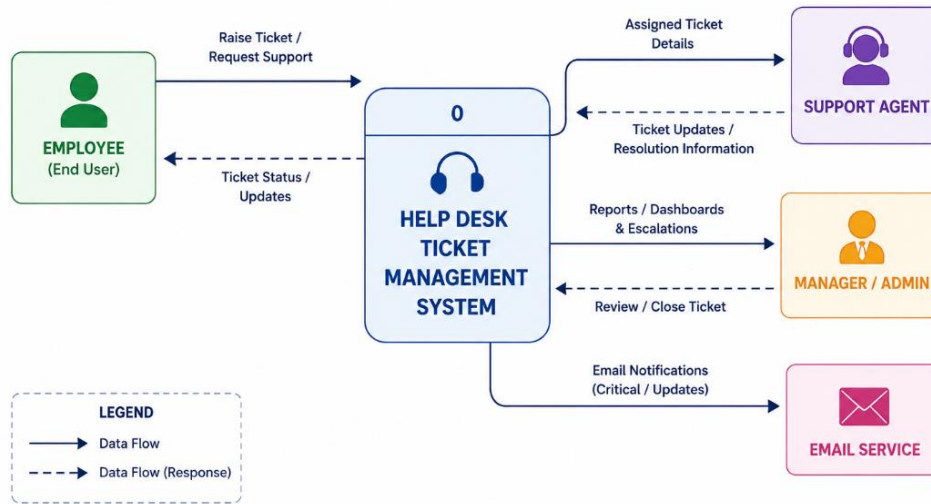
Non-functional Requirements

S.No	Non-functional Requirement	Description
1	Performance	The system should process ticket requests quickly and efficiently.
2	Security	Only authorized users can access the application based on assigned roles.
3	Availability	The system should be available whenever users need to raise or manage tickets.
4	Reliability	Ticket information should be stored accurately without data loss.
5	Scalability	The system should support increasing numbers of users and tickets.
6	Usability	The interface should be simple and user-friendly.
7	Maintainability	Future enhancements and modifications should be easy to implement.
8	Data Integrity	Ticket records should remain accurate and consistent throughout their lifecycle.
9	Compatibility	The application should work correctly in supported web browsers.
10	Auditability	The system should maintain logs of ticket activities for monitoring and reporting.

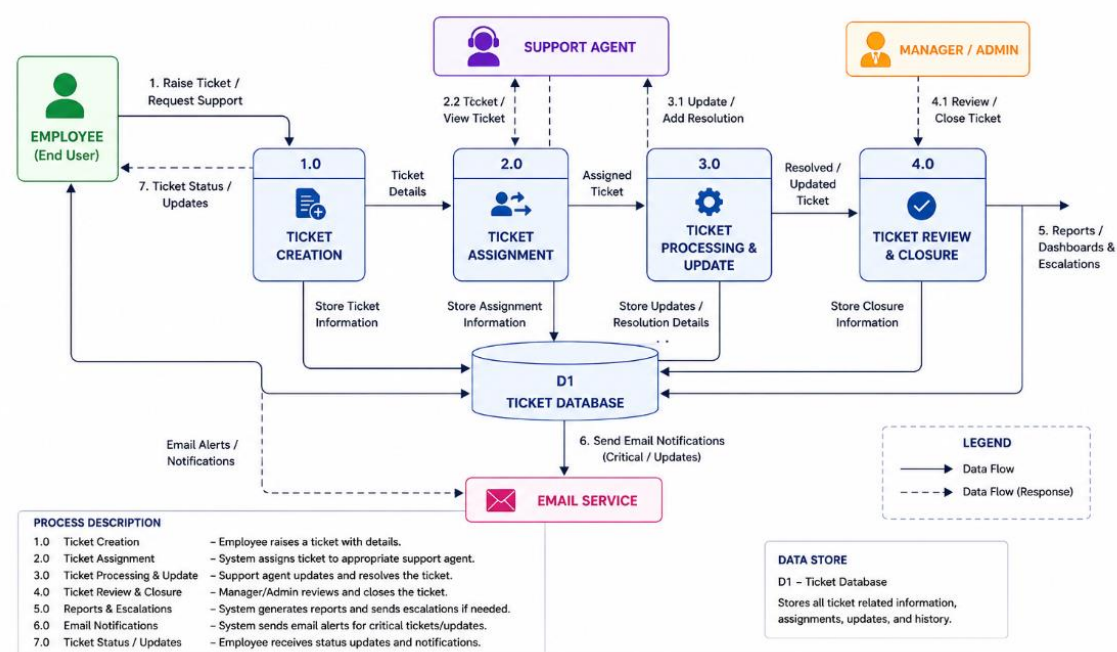
3.2 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of how information moves through a system. It illustrates the flow of data between users, processes, and data stores while showing how information is processed at each stage.

Data Flow Diagram LEVEL – 0



Data Flow Diagram LEVEL – 1



3.3 Technical Architecture

The Help Desk Ticket Management System using ServiceNow is built on the ServiceNow cloud platform. It provides role-based access for Employees, Support Agents, and Help Desk Managers. Business Rules and Flow Designer automate ticket assignment, status updates, and email notifications. Reports and Dashboards provide real-time monitoring of support activities.

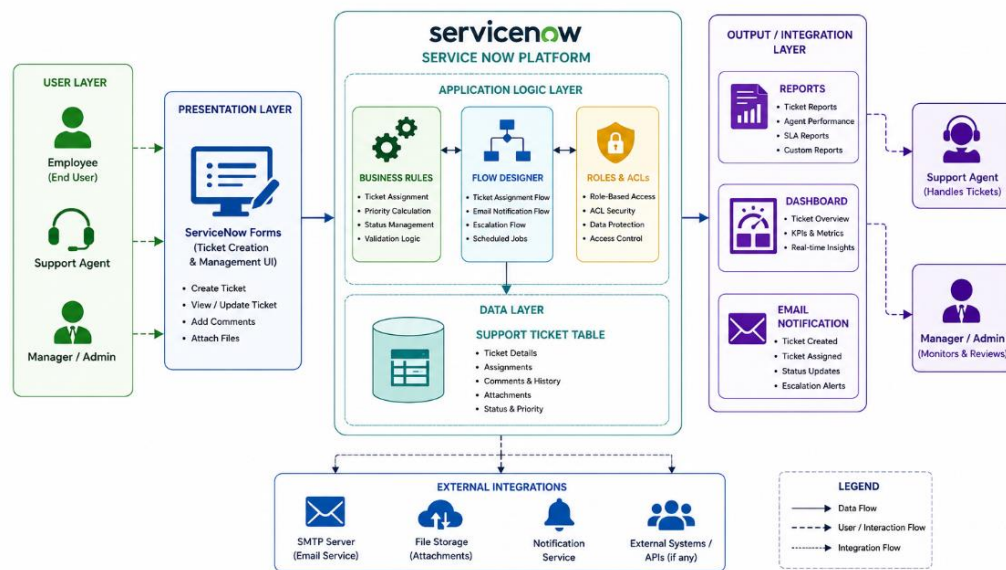


Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Ticket creation and management	ServiceNow Forms & Modules
2	Application Logic-1	Ticket assignment	Business Rules
3	Application Logic-2	Workflow automation	Flow Designer

4	Application Logic-3	Email notifications	Notification / Send Email
5	Database	Support ticket records	ServiceNow Tables
6	Cloud Platform	Application hosting	ServiceNow Cloud Platform
7	File Storage	Attachments for tickets	ServiceNow Attachments
8	Reporting	Generate reports	ServiceNow Reports
9	Dashboard	Visual analytics	ServiceNow Dashboards
10	Security	Role-based access	Roles & ACLs
11	Infrastructure	Cloud-hosted SaaS	ServiceNow

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Platform	Cloud-based application platform	ServiceNow
2	Security	Role-based access control	Roles, ACLs
3	Scalable Architecture	Supports multiple users and tickets	ServiceNow SaaS
4	Availability	Accessible through web browser	ServiceNow Cloud
5	Performance	Automated workflows and optimized data handling	Business Rules & Flow Designer

4. PROJECT DESIGN

4.1 Problem Solution Fit

- Employees can easily raise support tickets through the ServiceNow application.
- Business Rules automatically assign tickets to the appropriate support agent.
- Flow Designer automates workflows and email notifications for critical tickets.
- Employees can track the status of their tickets in real time.
- Support agents can efficiently update and resolve assigned tickets.
- Managers can review, monitor, and close support tickets.
- Reports and dashboards provide insights into ticket performance and support activities.
- Role-based access control ensures secure access for different user roles.
- The system reduces response time and improves issue resolution.
- A centralized ticket management platform enhances transparency, productivity, and overall support

Purpose:

- Identify the challenges faced by employees while requesting IT support.
- Understand the requirements of employees, support agents, and managers.
- Ensure the proposed solution effectively addresses the identified problems.
- Automate the ticket management process using ServiceNow.
- Reduce manual effort in ticket creation and assignment.
- Improve communication between employees and the IT support team.
- Enable real-time ticket tracking and status updates.
- Increase operational efficiency and user satisfaction.
- Support informed decision-making through reports and dashboards.
- Provide a secure and centralized platform for managing IT support requests.

Template:



4.2 Proposed Solution

S.No.	Parameter	Description
1	Problem Statement	Organizations often face delays in resolving IT support requests due to manual ticket handling, lack of centralized tracking, inefficient communication, and difficulty in monitoring ticket progress.
2	Solution Description	Develop a Help Desk Ticket Management System using ServiceNow that enables employees to raise support tickets, automatically assigns tickets to support agents using Business Rules, tracks ticket status, sends email notifications for critical issues, and provides reports and dashboards for effective monitoring.
3	Uniqueness	The solution integrates automated ticket assignment, role-based access control, real-time ticket tracking, critical email notifications, and interactive dashboards within a single ServiceNow application, reducing manual intervention and improving service efficiency.

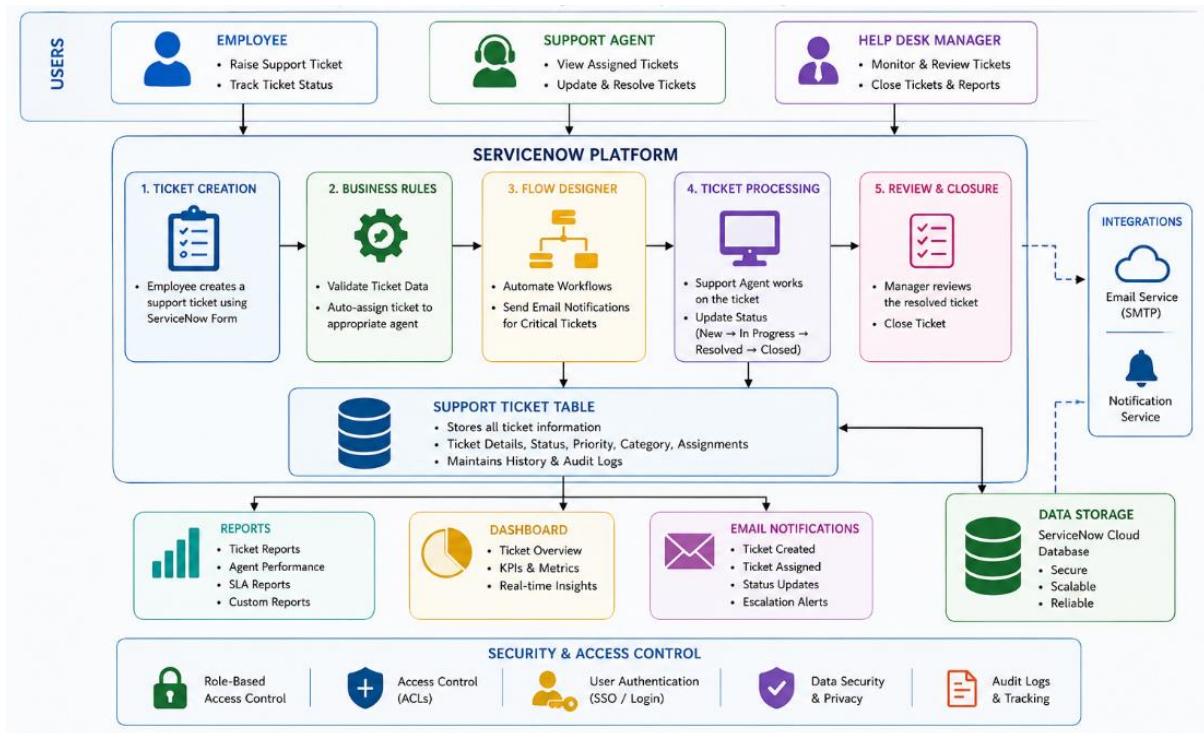
4	Customer Satisfaction	The system improves employee satisfaction by providing a transparent and efficient support process. Faster issue resolution, real-time status updates, and improved communication increase productivity and enhance the overall user experience.
5	Business Model (Revenue Model)	The solution can be offered as a cloud-based IT Service Management (ITSM) application for organizations using a subscription or Software-as-a-Service (SaaS) model. Organizations can customize and deploy the application according to their business requirements.
6	Scalability of the Solution	The application is built on the ServiceNow cloud platform, allowing it to scale easily with increasing users, departments, and support tickets. Additional workflows, modules, and integrations can be added without significant architectural changes.

4.3 Solution Architecture

The **Help Desk Ticket Management System using ServiceNow** is designed to automate and streamline the IT support process within an organization. The solution architecture bridges business requirements with ServiceNow technologies by integrating ticket creation, workflow automation, role-based access control, notifications, and reporting into a centralized cloud platform.

The proposed architecture ensures that support requests are processed efficiently while providing real-time visibility into ticket status, faster issue resolution, and secure access for different user roles.

Solution Architecture Diagram:



5. PROJECT PLANNING & SCHEDULING

5.1 Product Backlog

The Product Backlog is a prioritized list of all features, functionalities, and improvements required to develop the Help Desk Ticket Management System. It serves as the primary source of work for the project and is divided into user stories that are implemented across multiple sprints.

Product Backlog ID	Feature	Priority	Status

PB-01	Requirement Analysis	High	Completed
PB-02	Create Help Desk Application	High	Completed
PB-03	Create Support Ticket Table	High	Completed
PB-04	Design Forms & Modules	High	Completed
PB-05	Configure Roles & ACLs	High	Completed
PB-06	Implement Business Rules	High	Completed
PB-07	Configure Flow Designer	High	Completed
PB-08	Email Notification	Medium	Completed
PB-09	Reports	High	Completed
PB-10	Dashboard	High	Completed
PB-11	User Acceptance Testing	High	Completed
PB-12	Documentation	Medium	Completed

5.2 Sprint Planning

Sprint 1 (Foundation & Application Setup)

Epic 1 – Requirement Analysis

User Story	Description	Story Point
USN1	Analyze project requirements	2
USN2	Design project architecture	3

Epic 2 – Application Development

User Story	Description	Story Point
USN3	Create Help Desk application	2

USN4	Create Support Ticket table	3
USN5	Design ticket form and modules	3

Sprint 1 Story Points

$2 + 3 + 2 + 3 + 3 = 13$ **Story Points**

Sprint 2 (Automation & Security)

Epic 3 – Workflow Automation

User Story	Description	Story Point
USN6	Develop Business Rules	3
USN7	Configure Flow Designer	5
USN8	Configure Email Notifications	2

Epic 4 – Security & User Management

User Story	Description	Story Point
USN9	Create Roles and ACLs	5
USN10	Configure User Access	2

Sprint 2 Story Points

$3 + 5 + 2 + 5 + 2 = 17$ **Story Points**

Sprint 3 (Reporting & Testing)

Epic 5 – Analytics

User Story	Description	Story Point
------------	-------------	-------------

USN11	Create Reports	3
USN12	Develop Dashboard	5

Epic 6 – Testing & Documentation

User Story	Description	Story Point
USN13	Perform User Testing	2
USN14	Prepare Project Documentation	3

Sprint 3 Story Points

$3 + 5 + 2 + 3 = 13$ **Story Point**

Total Story Points

Sprint	Story Points
Sprint 1	13
Sprint 2	17
Sprint 3	13
Total	43

Final Statement

The project completed a total of 43 story points across three sprints, achieving an average velocity of approximately 14 story points per sprint. This sprint planning approach ensured systematic development, continuous progress tracking, and timely completion of the Help Desk Ticket Management System using ServiceNow.

5.3 Story Points

Story Points are used to estimate the effort required to complete each user story. The project follows the Fibonacci estimation technique (1, 2, 3, 5), where higher values indicate greater development effort.

Story Point	Effort Level
1	Very Easy
2	Easy
3	Moderate
5	Complex

Sprint	Story Points
Sprint 1	10
Sprint 2	15
Sprint 3	15
Total	40

Story Point Distribution

5.4 Velocity

Velocity Calculation

- Total Story Points = 40
- Number of Sprints = 3

Velocity = Total Story Points ÷ Number of Sprints

= $40 \div 3 = 13.33$

Average Velocity $\approx$ 13 Story Points per Sprint

If your faculty asks for story points per day:

- Total Duration = 6 Days
- Story Points = 40

Velocity per Day = $40 \div 6 = 6.67 \approx 7$ Story Points per Day

5.5 Burndown Chart

A Burndown Chart is a graphical representation of the remaining work against time during the project. It helps monitor sprint progress and ensures that development activities are completed within the planned schedule.

For the Help Desk Ticket Management System, the remaining story points decreased steadily as each sprint was completed.

Burndown Chart Interpretation

- Sprint 1: Initial development activities were completed, reducing the remaining work from 40 to 30 story points.
- Sprint 2: Core application features, automation, and security were implemented, reducing the remaining work to 15 story points.
- Sprint 3: Reports, dashboards, testing, and documentation were completed, reducing the remaining work to 0 story points and marking successful project completion.

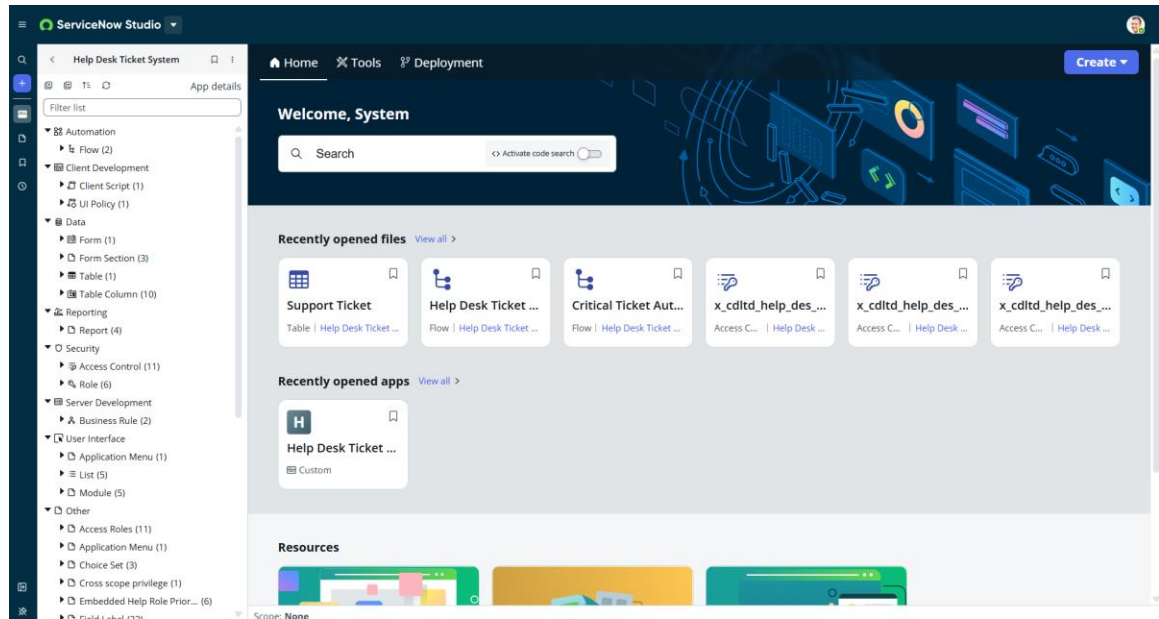
6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

1. Testing Scope

Module	Description
Ticket Creation	Create tickets
Business Rules	Auto assignment
Flow Designer	Critical workflow
Roles & ACLs	Security
Reports	Reporting
Dashboard	Analytics
Email Notification	Critical email

Application Home Page



2. Test Scenarios

ID	Scenario	Expected	Status
TS01	Create Ticket	Success	Pass
TS02	Auto Assignment	Assigned	Pass
TS03	Flow	Executed	Pass
TS04	Email	Generated	Pass
TS05	Reports	Displayed	Pass
TS06	Dashboard	Displayed	Pass

Ticket Form

servicenow

AllFavoritesHistoryWorkspaces

Support Ticket - Create HDT002043

Application scope: Help Desk Ticket System
Update set: Default [Help Desk Ticket System]

Ticket Information
New record

Submit

Ticket NumberHDT002043

Description

* Short Description

* Category-- None --

AssignmentStatus

* Requested By

Assigned To

Submit

Ticket List

servicenow

AllFavoritesHistoryWorkspacesAdmin

Support Tickets

Application scope: Help Desk Ticket System
Update set: Default [Help Desk Ticket System]

Support Tickets

Ticket Number

Search

Actions on selected rows...

New

All

	Ticket Number	Assigned To	Category	Description	Due Date	Priority	Requested By	Short Description	Status
	HDT002011	Abraham Lincoln	Hardware	setryui		Medium	Abel Tuter	asdfghj	New
	HDT002013	Alejandro Mascall	Network	asdfghjkl	2026-06-26 13:45:33	Medium	Alfonso Griglen	aszdxfghj00002	New
	HDT002014	(empty)	Hardware	zsxdcvbfghmnrj..0000003	2026-06-27 14:46:39	High	Alejandra Prenatt	asdfghmnj	Resolved
	HDT002015	Abraham Lincoln	Email	dfghjk		Medium	Alfonso Griglen	asdfghj	New
	HDT002016	(empty)	Other	asdfghjk		Medium	Abel Tuter	sdfghj	Resolved
	HDT002018	Aileen Mottern	Hardware	sdfghjkl;	2026-07-02 16:26:48	Medium	Alfonso Griglen	adsfghjkl	In Progress
	HDT002022	(empty)	Hardware			Low	Adela Cervantsz	asdfghjk	New
	HDT002024	(empty)	Network	qwertyuiop[]0000000000000000000000000000004	2026-06-27 06:41:20	Critical	Alejandra Prenatt	asdfghjkl;	New
	HDT002025	Software Support	Software	asfdghjkl222222222222		Medium	Joe Employee	asdfghmn,	New
	HDT002026	(empty)	Network	adsfghjkl333333333333		High	Lucius Best	sdfghjkl	New
	HDT002027	Hardware Support	Hardware	asdgfghghjk3333333333333		Medium	Adela Cervantsz	dfghj.k.	New
	HDT002028	Network Support	Network	asdfghjkl		Medium	Adela Cervantsz	asdfghj.	New
	HDT002029	Network Support	Network	qwertyuio []\789	2026-07-04 16:01:08	Critical	App Creation Admin	asdfghj	New

3. Defect Analysis

Resolution	Critical	High	Medium	Low	Total
Fixed	2	3	4	2	11
By Design	0	1	1	0	2
Duplicate	0	0	1	0	1
Not	0	0	0	1	1

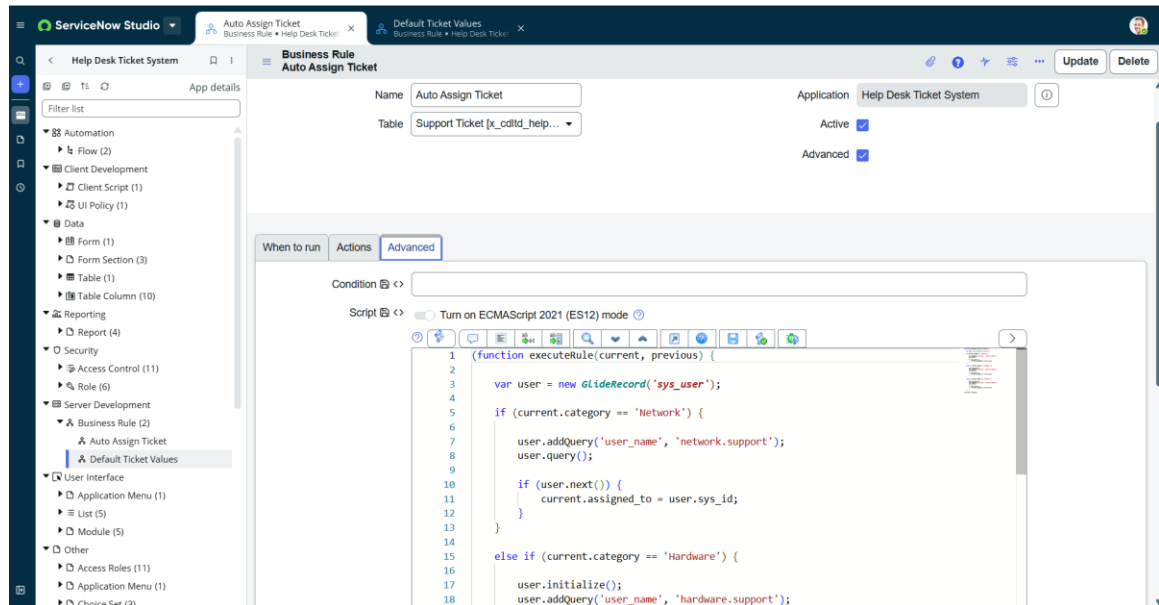
Reproduced					
Total	2	4	6	3	15

4. Test Case Analysis

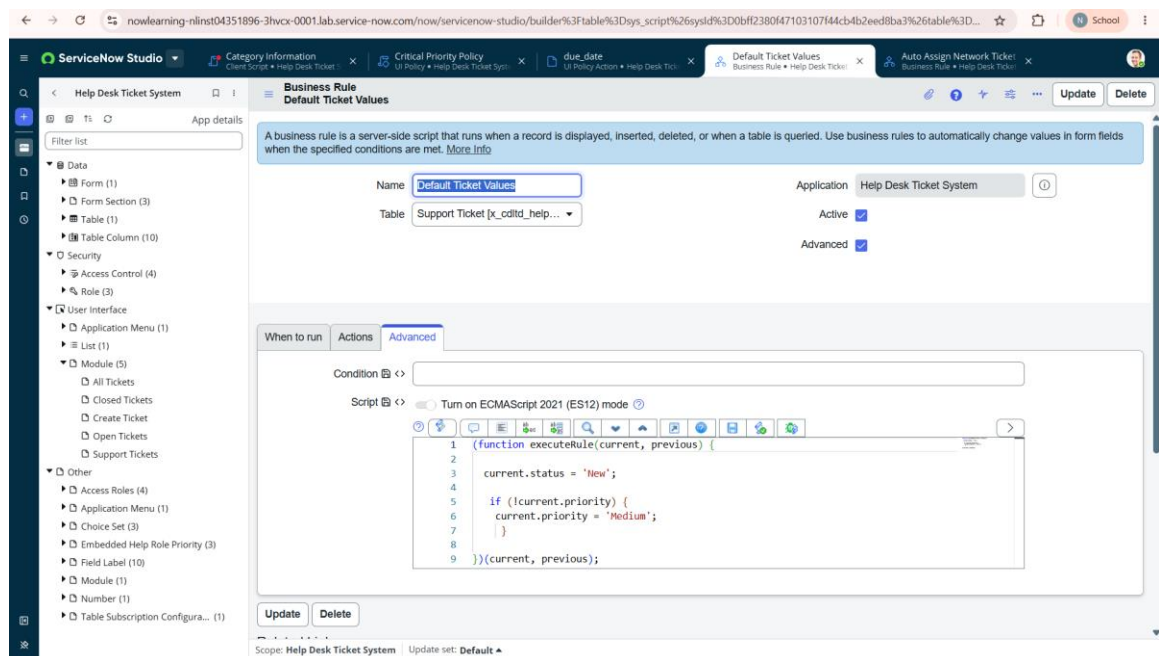
Module	Total	Not Tested	Failed	Passed
Ticket Creation	5	0	0	5
Business Rules	4	0	0	4
Flow Designer	4	0	0	4
Roles & ACLs	4	0	0	4
Email Notification	3	0	0	3
Reports	3	0	0	3
Dashboard	2	0	0	2
Ticket Lifecycle	4	0	0	4
Total	29	0	0	29

5. Screenshot Placeholders

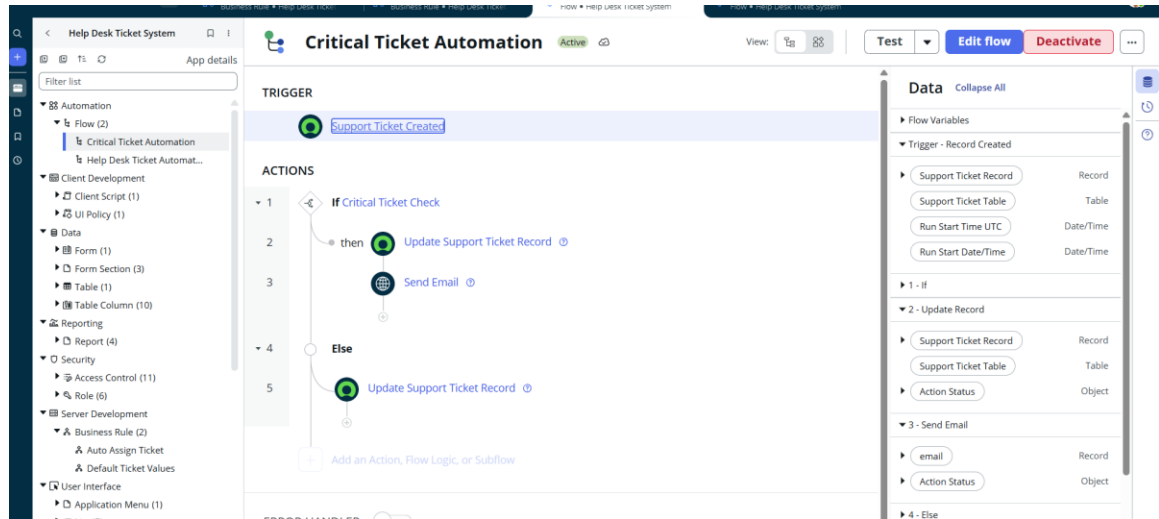
Business Rule- 1



Business Rule -2



Flow Designer



Assigned Ticket & Ticket Status

The screenshot shows the ServiceNow Support Tickets list view. The table displays a list of tickets with columns: Ticket Number, Assigned To, Category, Description, Due Date, Priority, Requested By, Short Description, and Status. Two tickets are visible: HDT002041 and HDT002042, both assigned to Nisha Employee and categorized as Hardware Support. The status for both tickets is "In Progress".

Ticket Number	Assigned To	Category	Description	Due Date	Priority	Requested By	Short Description	Status
HDT002041	Hardware Support	Hardware	zzzxsdcfvghj4120.	2026-07-04 08:59:45	Critical	Nisha Employee	laptop is not work for past 2 days	In Progress
HDT002042	Hardware Support	Hardware		2026-06-28 09:13:03	Critical	Nisha Employee	mouse pad is not working	In Progress

Outbound Email

The screenshot shows the ServiceNow Email - Critical Support Ticket Created preview window. The window displays the email body content, which includes a greeting, a message about a new Critical Support Ticket, and details about the ticket (Ticket Number: HDT002042, Priority: Critical, Category: Hardware, Requested By: e893bf04f47103107f44cb4b2eed8b4e). The email also includes a short description of the issue (mouse pad is not working) and a request to resolve the issue as soon as possible. The preview window is titled "Preview Email" and has a "Close" button.

Preview Email

Hello,

A new Critical Support Ticket has been created.

Ticket Number: HDT002042
Priority: Critical
Category: Hardware
Requested By: e893bf04f47103107f44cb4b2eed8b4e

Short Description: mouse pad is not working

Please resolve this issue as soon as possible.

Regards,
Help Desk Ticket System

Ref:MSG0000315_SQunzwHN5YW6jZlsJLFz

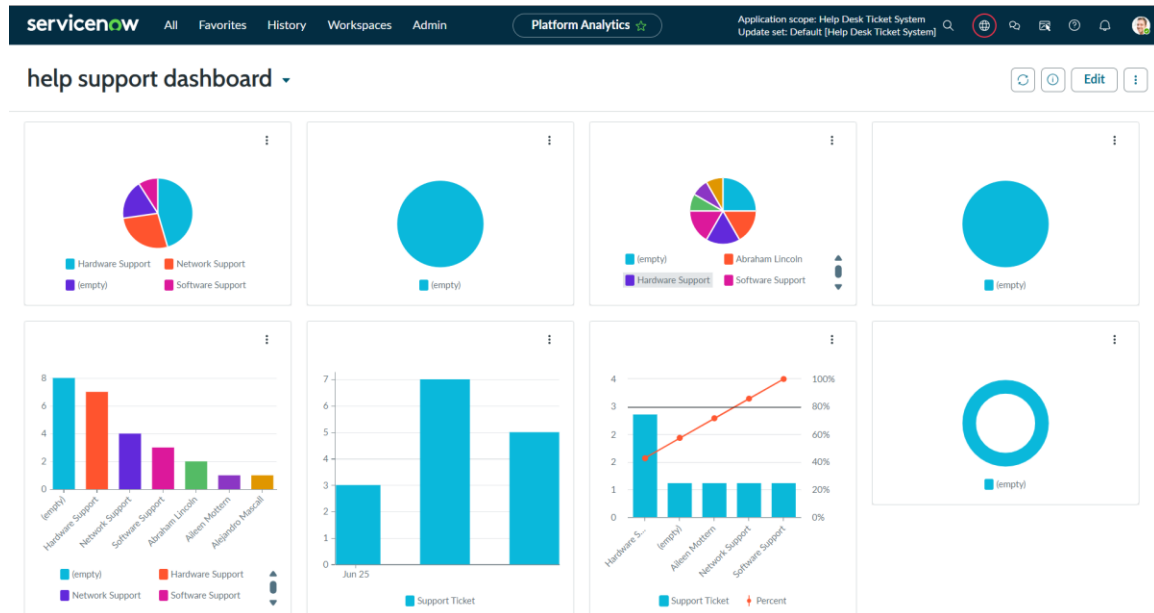
Reports

servicenow All Favorites History Workspaces Admin **Reports** Application scope: Help Desk Ticket System Update set: Default [Help Desk Ticket System]

≡ ⚙️ 📄 Reports Updated Search Actions on selected rows... New

	Title	Table	Type	Field Name	Created by	Updated
☒	Support Ticket by Priority	Support Ticket [x_cdtld_help_des_0_support_ticket]	Pie	assigned_to	admin	2026-06-27 07:00:54
☐	closed Ticket	Support Ticket [x_cdtld_help_des_0_support_ticket]	Bar	assigned_to	admin	2026-06-27 06:57:56
☐	Open Ticket	Support Ticket [x_cdtld_help_des_0_support_ticket]	Bar	assigned_to	admin	2026-06-27 06:56:39
☐	All Support Tickets	Support Ticket [x_cdtld_help_des_0_support_ticket]	List		admin	2026-06-27 06:53:36
☐	Usage of ServiceNow Applications	Usage Data for Applications [ua_app_usage]	Bar	app_name	admin	2026-05-27 20:00:04
☐	CMDB Compliance Result List	CMDB Health Result [cmdb_health_result]	List		admin	2026-05-27 20:00:04

Dashboard



7. RESULTS

7.1 Output Screenshots

Insert:

Step 1 : Application Menu

Navigation:

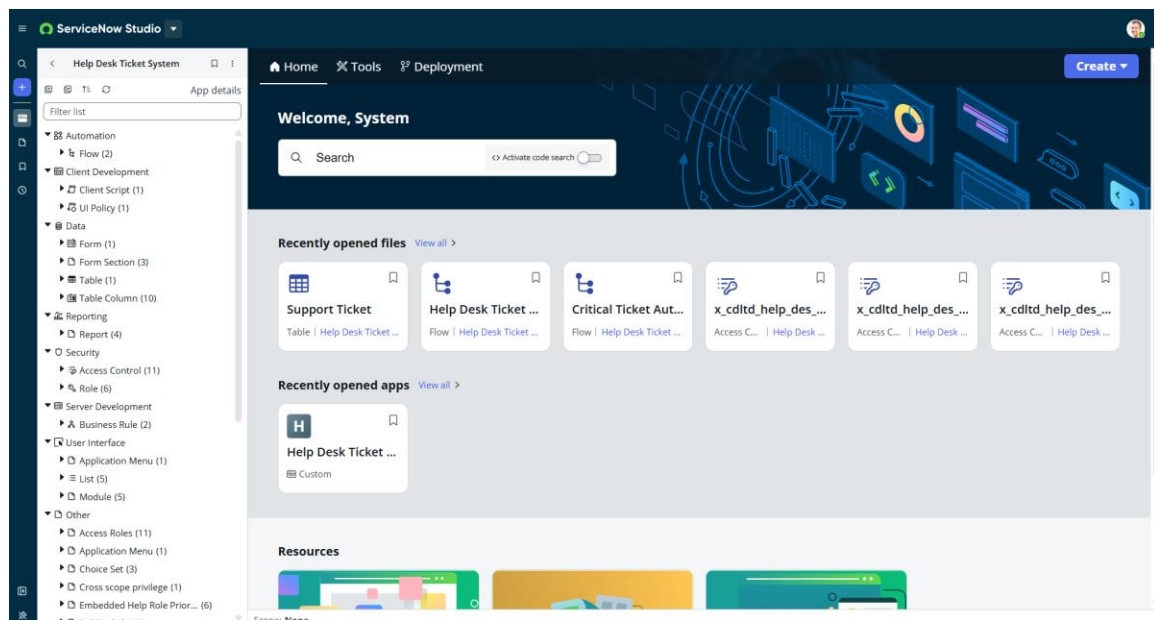
All

→ ServiceNow Studio

→ Create Application

→ Name: Help Desk Tickets Management System

→ Create



Step 2 :Create Support Ticket Table

Navigation:

ServiceNow Studio

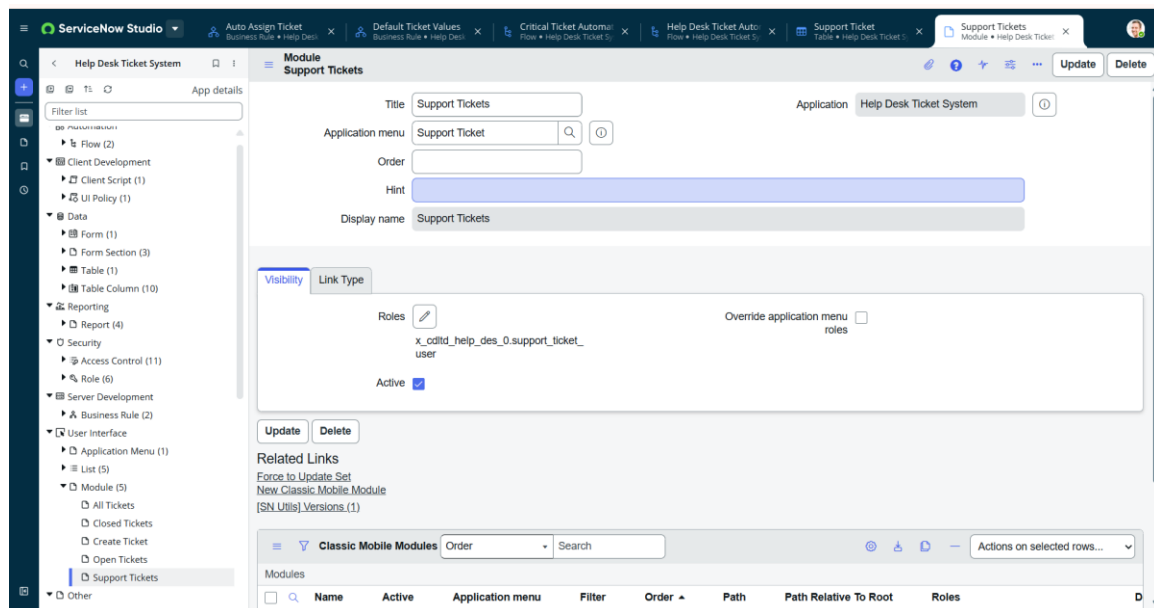
→ Help Desk Tickets Management System

→ Create Application File

→ Table

→ Name: Support Ticket

→ Create



Step 3 :Create Table Fields

Navigation:

ServiceNow Studio

→ Support Ticket Table → Columns → New

ServiceNow Studio

Support Ticket

Ticket Information

Ticket Number: HDT002044

Description: [Empty]

Short Description: [Empty]

Category: -- None --

Assignment

Requested By: [Empty]

Assigned To: [Empty]

Status

Priority: Medium

Status: New

[Open form in Platform](#)

Step 4: Create Application Modules

Navigation:

ServiceNow Studio

→ Create Application File

→ Module

1. Create Ticket

ServiceNow Studio

Module: Create Ticket

Title: Create Ticket

Application: Help Desk Ticket System

Application menu: [Empty]

Order: [Empty]

Hint: [Empty]

Display name: Create Ticket

Link Type

Select the type of link for the module or select separator to create a horizontal line. The fields below change depending on your selection.

Link type: List of Records

* Table: Support Ticket [x_cdtid_help_des_0_support_ticket]

View name: [Empty]

Filter: Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Arguments: [Empty]

2. Open Tickets

The screenshot shows the ServiceNow Studio interface for configuring the 'Open Tickets' module. The left sidebar displays a tree view of the application structure, including Data, Security, User Interface, and Other sections. The main workspace is titled 'Module Open Tickets' and contains the following fields:

- Title:** Open Tickets
- Application menu:** (empty)
- Order:** (empty)
- Hint:** (empty)
- Display name:** Open Tickets

Below these fields, the 'Link Type' tab is selected. It contains a message: 'Select the type of link for the module or select separator to create a horizontal line. The fields below change depending on your selection.' The configuration fields are:

- Link type:** List of Records
- * Table:** Support Ticket [x_cdtid_help_des_0_support_ticket]
- View name:** (empty)
- Filter:** Add Filter Condition, Add OR Clause
- Arguments:** (empty)
- Window name:** (empty)

3.All Tickets

The screenshot shows the ServiceNow Studio interface for configuring the 'All Tickets' module. The left sidebar displays a tree view of the application structure, including Data, Security, User Interface, and Other sections. The main workspace is titled 'Module All Tickets' and contains the following fields:

- Title:** All Tickets
- Application menu:** (empty)
- Order:** (empty)
- Hint:** (empty)
- Display name:** All Tickets

Below these fields, the 'Link Type' tab is selected. It contains a message: 'Select the type of link for the module or select separator to create a horizontal line. The fields below change depending on your selection.' The configuration fields are:

- Link type:** List of Records
- * Table:** Support Ticket [x_cdtid_help_des_0_support_ticket]
- View name:** (empty)
- Filter:** Add Filter Condition, Add OR Clause
- Arguments:** (empty)
- Window name:** (empty)

4.Closed Tickets

ServiceNow Studio - Help Desk Ticket System

Module Closed Tickets

Title:

Application:

Application menu:

Order:

Hint:

Display name:

Link Type

Select the type of link for the module or select separator to create a horizontal line. The fields below change depending on your selection.

Link type:

* Table:

View name:

Filter:

-- choose field -- -- oper -- -- value --

Arguments:

Window name:

Step 5: Create Roles

Navigation:

All

→ User Administration

→ Roles

→ New

Name	Description	Elevated privilege
x_cdtld_help_des_0.admin	Default admin role	false
x_cdtld_help_des_0.helpdesk_agent		false
x_cdtld_help_des_0.helpdesk_manager		false
x_cdtld_help_des_0.helpdesk_user		false
x_cdtld_help_des_0.support_ticket_user		false
x_cdtld_help_des_0.user	Default user role	false

Step 6: Configure ACLs

Navigation:

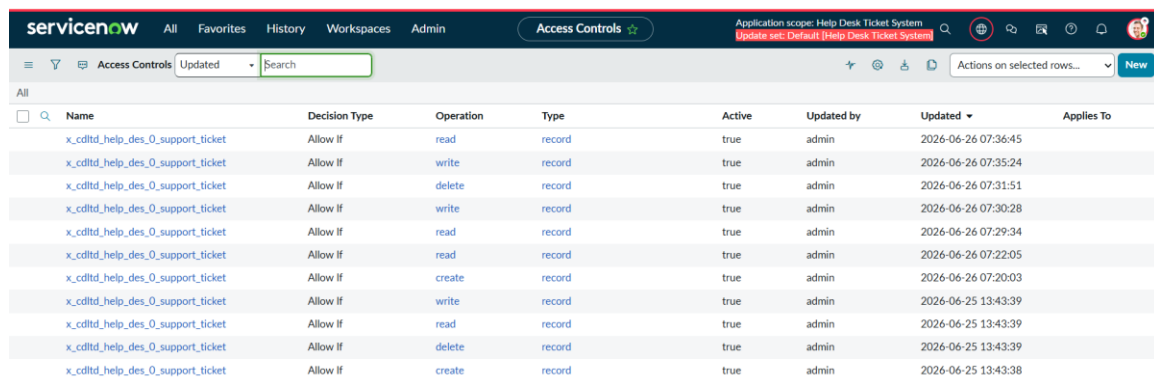
Elevate Role

All

→ System Security

→ Access Control (ACL)

→ New



The screenshot shows the ServiceNow 'Access Controls' table. The table has columns for Name, Decision Type, Operation, Type, Active, Updated by, Updated, and Applies To. It lists 12 records for the 'x_cdttd_help_des_0_support_ticket' table, all with 'Allow If' decision type and 'record' type. Operations include read, write, delete, and create. All records are active and were updated by 'admin'.

Name	Decision Type	Operation	Type	Active	Updated by	Updated	Applies To
x_cdttd_help_des_0_support_ticket	Allow If	read	record	true	admin	2026-06-26 07:36:45	
x_cdttd_help_des_0_support_ticket	Allow If	write	record	true	admin	2026-06-26 07:35:24	
x_cdttd_help_des_0_support_ticket	Allow If	delete	record	true	admin	2026-06-26 07:31:51	
x_cdttd_help_des_0_support_ticket	Allow If	write	record	true	admin	2026-06-26 07:30:28	
x_cdttd_help_des_0_support_ticket	Allow If	read	record	true	admin	2026-06-26 07:29:34	
x_cdttd_help_des_0_support_ticket	Allow If	read	record	true	admin	2026-06-26 07:22:05	
x_cdttd_help_des_0_support_ticket	Allow If	create	record	true	admin	2026-06-26 07:20:03	
x_cdttd_help_des_0_support_ticket	Allow If	write	record	true	admin	2026-06-25 13:43:39	
x_cdttd_help_des_0_support_ticket	Allow If	read	record	true	admin	2026-06-25 13:43:39	
x_cdttd_help_des_0_support_ticket	Allow If	delete	record	true	admin	2026-06-25 13:43:39	
x_cdttd_help_des_0_support_ticket	Allow If	create	record	true	admin	2026-06-25 13:43:38	

Step 7: Create Business Rule

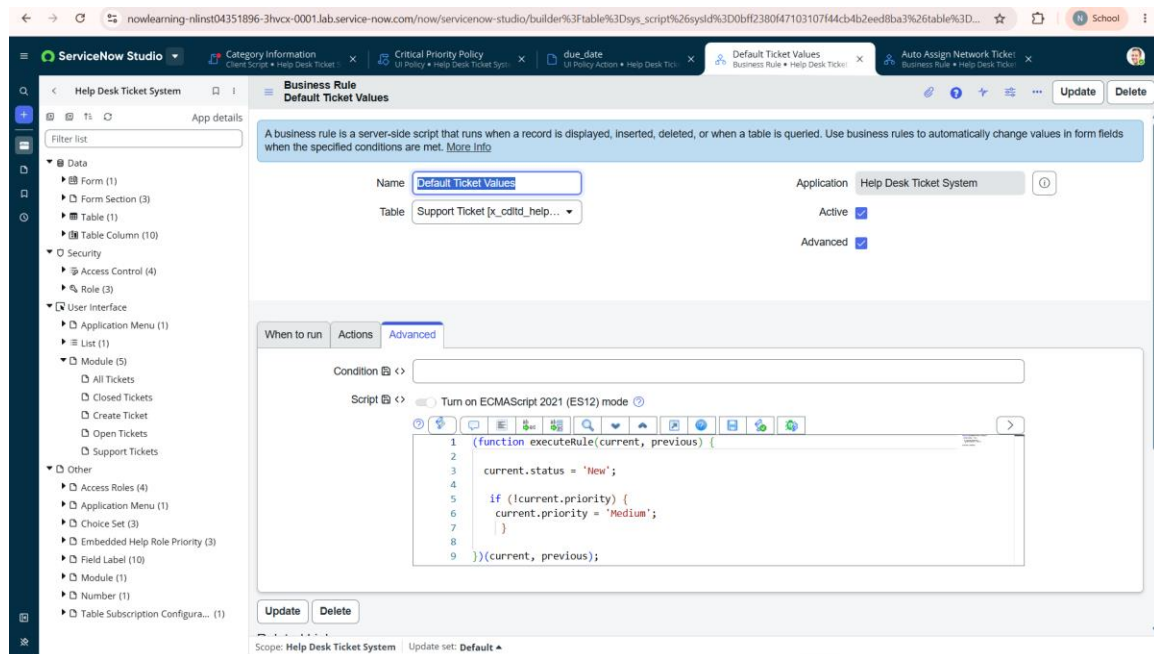
Navigation:

ServiceNow Studio

→ Create Application File

→ Business Rules

→ New



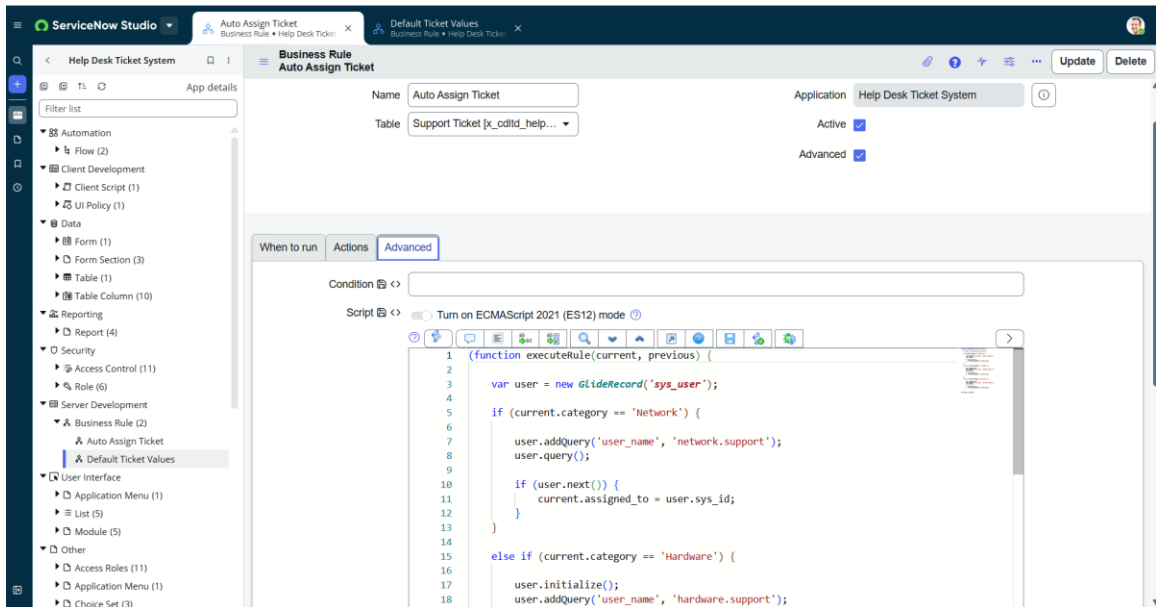
The screenshot shows the ServiceNow Studio interface for creating a business rule. The left sidebar shows a tree view of the application structure. The main area is titled 'Business Rule' and 'Default Ticket Values'. It includes fields for Name, Table, Application, Active, and Advanced. The 'Advanced' tab is selected, showing a script editor with ECMAScript 2021 (ES12) mode. The script defines a function 'executerRule' that sets the status to 'New' and updates the priority to 'Medium' if the current priority is not 'Medium'.

Business Rule Details:

- Name: Default Ticket Values
- Table: Support Ticket [x_cdttd_help_...]
- Application: Help Desk Ticket System
- Active: ☒
- Advanced: ☒

Script (ES12 mode):

```
1 function executerRule(current, previous) {  
2  
3   current.status = 'New';  
4  
5   if (!current.priority) {  
6     current.priority = 'Medium';  
7   }  
8  
9 }(current, previous);
```



Step 8: Create Flow Designer

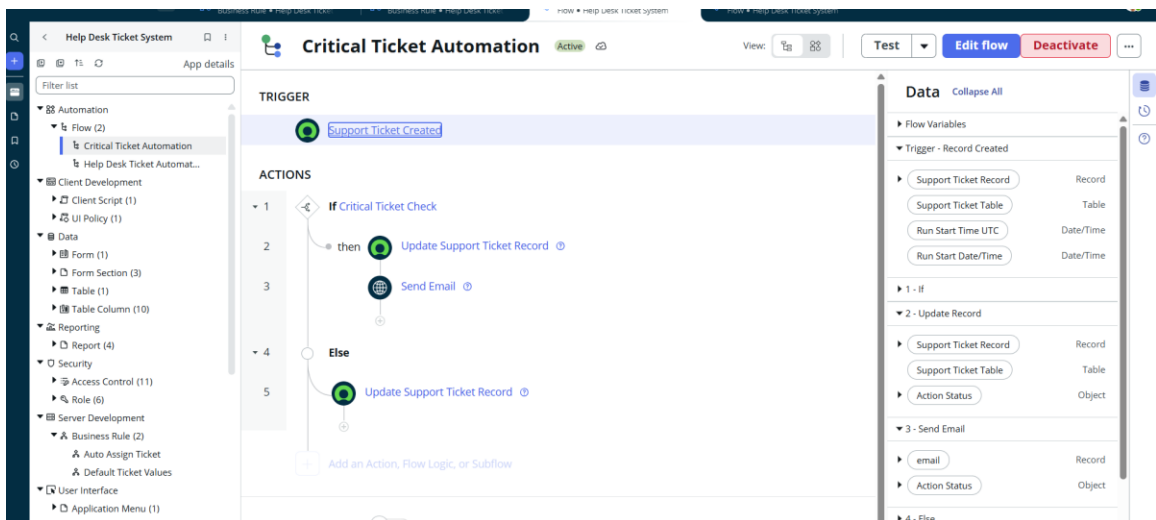
Navigation:

All

→ Process Automation

→ Flow Designer

→ New



Flow:

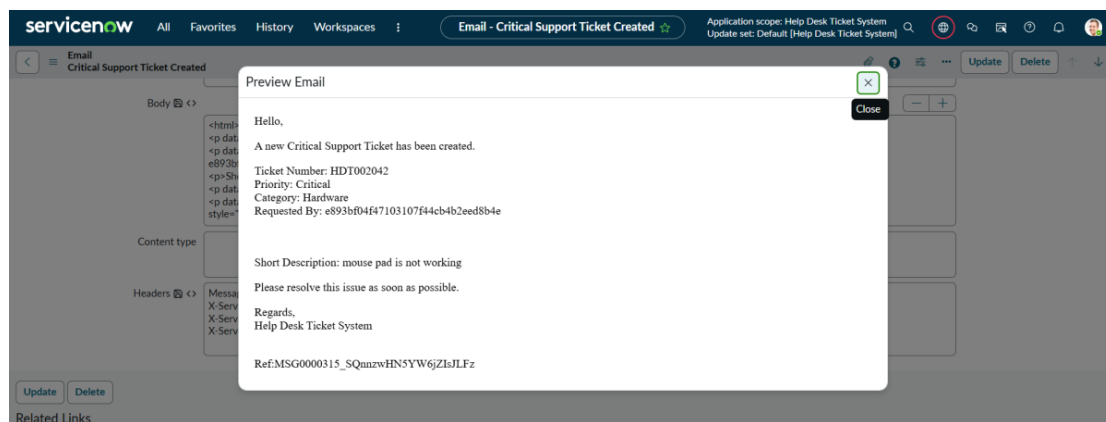
- Trigger → Ticket Created
- Condition → Priority = Critical
- Action → Send Email Notification

Step 9: Create Email Notification

Navigation:

All

→ System Notification → Email → Notifications → New



Step 10: Create Reports

Navigation:

All

→ Reports

→ Create New

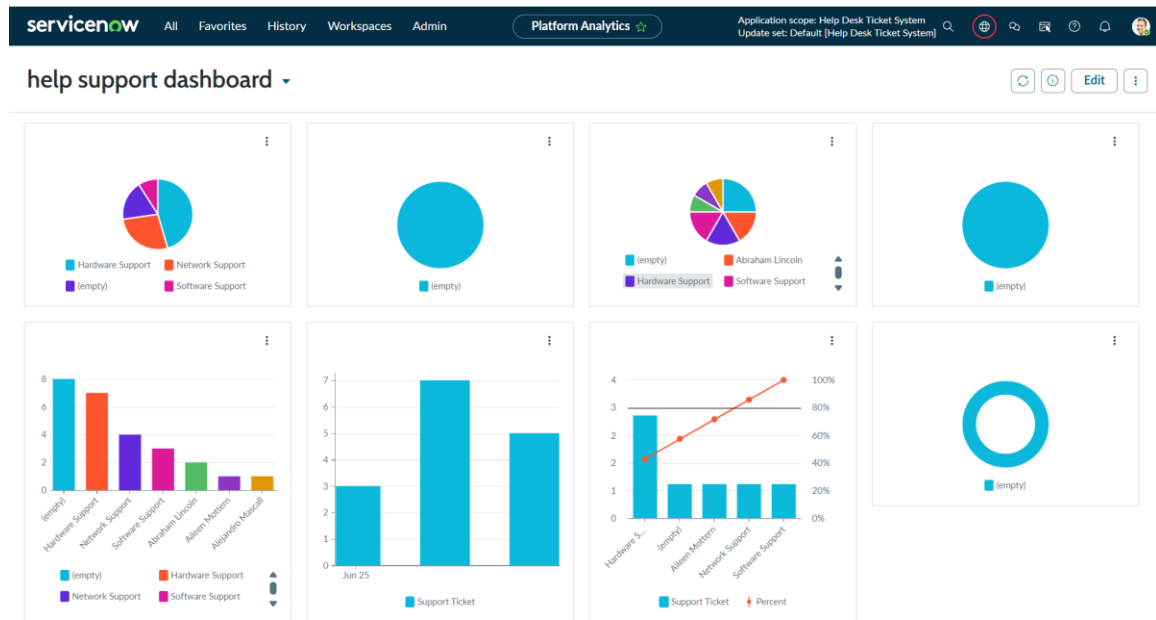
Title	Table	Type	Field Name	Created by	Updated
Support Ticket by Priority	Support Ticket [x_cdltd_help_des_0_support_ticket]	Pie	assigned_to	admin	2026-06-27 07:00:54
closed Ticket	Support Ticket [x_cdltd_help_des_0_support_ticket]	Bar	assigned_to	admin	2026-06-27 06:57:56
Open Ticket	Support Ticket [x_cdltd_help_des_0_support_ticket]	Bar	assigned_to	admin	2026-06-27 06:56:39
All Support Tickets	Support Ticket [x_cdltd_help_des_0_support_ticket]	List		admin	2026-06-27 06:53:36
Usage of ServiceNow Applications	Usage Data for Applications [ua_app_usage]	Bar	app_name	admin	2026-05-27 20:00:04
CMDB Compliance Result List	CMDB Health Result [cmdb_health_result]	List		admin	2026-05-27 20:00:04

Step 11: Create Dashboard

Navigation:

All

→ Self-Service → Dashboards → New Dashboard



Step 12: Test as Employee

Navigation:

User Menu

→ Impersonate User

→ Employee

Then:

Support Ticket

→ Create Ticket

→ Submit

- Employee Ticket

servicenow All Favorites History Workspaces Admin Support Tickets Application scope: Help Desk Ticket System Update set: Default [Help Desk Ticket System]									
Support Tickets Ticket Number Search Actions on selected rows... New									
All > Status = New .or. Status = In Progress .or. Status = On Hold									
Ticket Number	Assigned To	Category	Description	Due Date	Priority	Requested By	Short Description	Status	
HDT002041	Hardware Support	Hardware	zzxsdcfvghj4120.	2026-07-04 08:59:45	Critical	Nisha Employee	laptop is not work for past 2 days	In Progress	
HDT002042	Hardware Support	Hardware		2026-06-28 09:13:03	Critical	Nisha Employee	mouse pad is not working	In Progress	

• Assigned Ticket

servicenow All Favorites History Workspaces Admin Support Tickets Application scope: Help Desk Ticket System Update set: Default [Help Desk Ticket System]									
Support Tickets Ticket Number Search Actions on selected rows... New									
All > Status = New .or. Status = In Progress .or. Status = On Hold									
Ticket Number	Assigned To	Category	Description	Due Date	Priority	Requested By	Short Description	Status	
HDT002041	Hardware Support	Hardware	zzxsdcfvghj4120.	2026-07-04 08:59:45	Critical	Nisha Employee	laptop is not work for past 2 days	In Progress	
HDT002042	Hardware Support	Hardware		2026-06-28 09:13:03	Critical	Nisha Employee	mouse pad is not working	In Progress	

• Ticket Status

servicenow All Favorites History Workspaces Admin Support Tickets Application scope: Help Desk Ticket System Update set: Default [Help Desk Ticket System]									
Support Tickets Ticket Number Search Actions on selected rows... New									
All > Status = New .or. Status = In Progress .or. Status = On Hold									
Ticket Number	Assigned To	Category	Description	Due Date	Priority	Requested By	Short Description	Status	
HDT002041	Hardware Support	Hardware	zzxsdcfvghj4120.	2026-07-04 08:59:45	Critical	Nisha Employee	laptop is not work for past 2 days	In Progress	
HDT002042	Hardware Support	Hardware		2026-06-28 09:13:03	Critical	Nisha Employee	mouse pad is not working	In Progress	

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Automated workflows
- Faster ticket resolution
- Secure role-based access
- Real-time reports

Disadvantages:

- Requires ServiceNow instance
- Internet dependency

9. Conclusion

- The Help Desk Ticket Management System using ServiceNow successfully automates the complete ticket management lifecycle, from ticket creation to closure.

- The system reduces manual effort by implementing automated ticket assignment, workflow management, and role-based access control.
- Employees can easily raise and track support requests, while support agents and managers can efficiently manage and resolve tickets.
- Features such as Business Rules, Flow Designer, Email Notifications, Reports, and Dashboards improve communication, transparency, and operational efficiency.
- The project demonstrates how ServiceNow can be effectively used to develop secure, scalable, and user-friendly enterprise applications for IT service management.
- Overall, the developed solution enhances productivity, reduces response time, and provides a reliable platform for managing organizational support requests.

10. FUTURE SCOPE

- AI-powered ticket classification
- Chatbot integration
- SLA monitoring
- Mobile application
- Third-party integrations

11. APPENDIX

Source Code:

Business Rule for Auto Assign Ticket

```
(function executeRule(current, previous) {
    var user = new GlideRecord('sys_user');
    if (current.category == 'Network') {
        user.addQuery('user_name', 'network.support');
        user.query();
        if (user.next()) {
            current.assigned_to = user.sys_id;
        }
    }
}
```

```

    }
    else if (current.category == 'Hardware') {
        user.initialize();
        user.addQuery('user_name', 'hardware.support');
        user.query();
        if (user.next()) {
            current.assigned_to = user.sys_id;
        }
    }
    else if (current.category == 'Software') {
        user.initialize();
        user.addQuery('user_name', 'software.support');
        user.query();
        if (user.next()) {
            current.assigned_to = user.sys_id;
        }
    }
})(current, previous);

```

Business Rule For Default Assign Ticket

```

(function executeRule(current, previous) {
    if(!current.status){
        current.status = 'New';
    }
    if (!current.priority) {
        current.priority = 'Medium';
    }
}

```

```
}  
})(current, previous);
```

GitHub Repository:

Project Demo Link:

https://drive.google.com/file/d/1FsArDAHfy8DO_y9wIIsp8OGvHnE4DHWf/view?usp=sharing